ask, but it turned out that the intelligent questioning approach managed to be faster and more accurate.

The trials also showed the system to be more intelligent than it was expected to be, because it could even understand complex instructions with lots of prepositions. For example, it could respond accurately when somebody said, "Hand me the spoon to the left of the bowl." Although such complex phrases were not built into the language model, the robot was able to use intelligent social feedback to figure out the instruction.

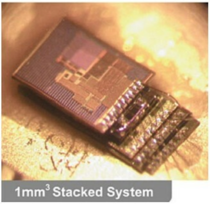
Learning gets deeper and smaller than you thought

Deep learning is an artificial intelligence (AI) technology that is pervading all streams of life ranging from banking to baking. A deep learning system essentially uses neural networks, modelled after the human brain, to learn by itself just like a human child does. It is made of multi-layered deep neural networks that mimic the activities of the layers of neurons in the neocortex. Each layer tries to understand something more than the previous layer, thereby developing a deeper understanding of things. The resulting system is self-learning, which means that it is not restricted by what it has been taught to do. It can react according to the situation and even make decisions by itself.

Deep learning is obviously a very useful tech for robots, too. However, it usually requires large memory banks and runs on huge servers powered by advanced graphics processing units (GPUs). If only deep learning could be achieved in a form factor small enough to embed in a robot!

Micromotes developed at University of Michigan, USA, could be the answer to this challenge. Measuring one cubic millimetre, the micromotes developed by David Blaauw and his colleague Dennis Sylvester are amongst the world's smallest computers. The duo has developed different variants of micromotes, including smart sensors and radios. Amongst these is a micromote that incorporates a deep learning processor, which can operate a neural network using just 288 microwatts.

There have been earlier attempts to reduce the size and power demands of deep learning using dedicated hardware specially designed to run these algorithms. But so far, nobody has managed to use less than 50 milliwatts of power and the size too has never been this small. Blaauw



Tiny micromotes developed at University of Michigan can incorporate deep learning processors in them (Image courtesy: University of Michigan)

and team managed to achieve deep learning on a micromote by redesigning the chip architecture, with tweaks such as situating four processing elements within the memory (SRAM) to minimise data movement.

The team's intention was to bring deep learning to the Internet of Things (IoT), so we can have devices like security cameras with onboard deep learning processors that can instantly differentiate between a branch and a thief lurking on the tree. But the same technology can be very useful for robots, too.

A hardware-agnostic approach to deep learning

Max Versace's approach to low-power AI for robots is a bit different. Versace's idea can be traced back to 2010, when NASA approached him and his team with the challenge of developing a software controller for